PARISHRAM 2.0

CLASS 12TH 2026

MATRIX

MATHS **LECTURE-2**

BY-MANDAR SIR



TOPICS TO BE COVERED

1. Questions on Addition & Subtraction
2. Pollwa Questions → Ready
3. Transpose of Matrix
4. Symmetric & Skew Symmetric

Why Fear When Rolex Sir is Here



Scalar Multiplication of Matrix

$$A = \begin{bmatrix} 7 & 4 \\ 3 & 5 \end{bmatrix}$$

$$5A = 5\begin{bmatrix} 7 & 4 \\ 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5\times7 & 5\times4 \\ 5\times3 & 5\times5 \end{bmatrix}$$

$$= \begin{bmatrix} 35 & 20 \\ 15 & 25 \end{bmatrix}$$

$$B = \begin{bmatrix} 7 & 2 & 1 \\ -1 & 0 & 10 \end{bmatrix}$$

$$4B = 4\begin{bmatrix} 7 & 2 & 1 \\ -1 & 0 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 4\times7 & 4\times2 & 4\times1 \\ 4\times-1 & 4\times0 & 4\times10 \end{bmatrix}$$

$$= \begin{bmatrix} 28 & 8 & 4 \\ -4 & 0 & 40 \end{bmatrix}$$

QUESTION

$$\begin{bmatrix}\cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha\end{bmatrix} + \begin{bmatrix}\cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix}$$



Find α

$$\begin{bmatrix}2\cos\alpha & -\sin\alpha + \sin\alpha \\ \sin\alpha + (-\sin\alpha) & 2\cos\alpha\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}2\cos\alpha & 0 \\ 0 & 2\cos\alpha\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix}$$

2×2 $\qquad$ 2×2

$2\cos\alpha = 1$

$\cos\alpha = \frac{1}{2}$

$\alpha = \cos^{-1}\left(\frac{1}{2}\right)$

$\alpha = \pi/3$

POLL QUESTION

$x \times y$

If $\begin{bmatrix} x+y & 7 \\ 9 & x-y \end{bmatrix} = \begin{bmatrix} 2 & 7 \\ 9 & 4 \end{bmatrix}$, then $x \cdot y$ is equal to _____.



1. 3
2. 2
3. −3
4. −2

$x + y = 2$ — (i)

$x - y = 4$ — (ii)

$2x = 6$

$2x = 6$

$x = 6/2$

$x = 3$

Put $x = 3$ in eq (i)

$3 + y = 2$

$y = 2 - 3$

$y = -1$

$xy = 3(-1)$

$= -3$

POLL QUESTION

If $3A - B = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, then find the matrix A is



1. $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$
2. $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ ✓
3. $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$
4. $\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$

$3A - \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix}$

$3A = \begin{bmatrix} 5 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$

$3A = \begin{bmatrix} 5+4 & 0+3 \\ 1+2 & 1+5 \end{bmatrix}$

$3A = \begin{bmatrix} 9 & 3 \\ 3 & 6 \end{bmatrix}$

$A = \frac{1}{3}\begin{bmatrix} 9 & 3 \\ 3 & 6 \end{bmatrix}$

$A = \begin{bmatrix} \frac{1}{3}\times 9 & \frac{1}{3}\times 3 \\ \frac{1}{3}\times 3 & \frac{1}{3}\times 6 \end{bmatrix}$

$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

POLL QUESTION

If $A + B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $A - 2B = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, then A is equal to _____.



1. $\frac{1}{3}\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$
2. $\frac{1}{3}\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$
3. $\frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$
4. $\frac{1}{3}\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$

$A = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$

$A + B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ —(i)

$A - 2B = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ —(ii)

Multiply by 2 in eq (i)

$2A + 2B = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}$ —(iii)

Add (ii) & (iii)

$2A + 2B + A - 2B = \begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$

$3A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$

POLL QUESTION

Which of the given values of x and y make the following pair of matrices equal?

$$\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix} = \begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix}$$



1. $x = \frac{-1}{3}, y = 7$
2. Not possible
3. $y = 7, x = \frac{-2}{3}$
4. $x = \frac{-1}{3}, y = \frac{-2}{3}$

x and y

$y + 1 = 8$
$y = 8 - 1$
$y = 7$

$3x + 7 = 0$
$3x = -7$
$x = -7/3$

$5 = y - 2$
$5 + 2 = y$
$y = 7$

$2 - 3x = 4$
$2 - 4 = 3x$
$-2 = 3x$
$-2/3 = x$

Not possible

POLL QUESTION

If $\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$, then value of $a + b - c + 2d$ is



1. 8
2. 10
3. 4
4. -8

$2a+b = 4$ —(i)

$a-2b = -3$ —(ii)

Multiply eq(i) by 2

$4a+2b = 8$ —(iii)

$a-2b = -3$

$5a = 5$

$a = 5/5$

$\boxed{a = 1}$

Put $a = 1$ in eq(i)

$2(1)+b = 4$

$2+b = 4$

$b = 4-2$

$\boxed{b = 2}$

$5c-d = 11$ —(iv)

$4c+3d = 24$ —(v)

$5c - d = 11 \quad -(iv)$

$4c + 3d = 24 \quad -(v)$

Multiply eq (iv) by 3

$15c - 3d = 33$

$4c + 3d = 24$

$19c = 57$

$c = 57/19$

$\boxed{c = 3}$

Put $c = 3$ in eq (iv)

$5(3) - d = 11$

$15 - d = 11$

$15 - 11 = d$

$\boxed{4 = d}$

$a + b - c + 2d$

$= 1 + 2 - 3 + 2(4)$

$= 3 - 3 + 8$

$\rightarrow 8$

QUESTION

Find non-zero values of x satisfying the matrix equation

$$x\begin{bmatrix}2x & 2\\ 3 & x\end{bmatrix}+2\begin{bmatrix}8 & 5x\\ 4 & 4x\end{bmatrix}=2\begin{bmatrix}(x^2+8) & 24\\ 10 & 6x\end{bmatrix}$$

$x^2+8x=12x$

$x^2+8x-12x=0$

$x^2-4x=0$

$x(x-4)=0$

$x=0$ ✗ or $x-4=0$

$x=4$

$$\begin{bmatrix}2x^2 & 2x\\ 3x & x^2\end{bmatrix}+\begin{bmatrix}16 & 10x\\ 8 & 8x\end{bmatrix}=\begin{bmatrix}2x^2+16 & 48\\ 20 & 12x\end{bmatrix}$$

$$\begin{bmatrix}2x^2+16\\ 3x+8\end{bmatrix}\quad\begin{bmatrix}2x+10x\\ x^2+8x\end{bmatrix}=\begin{bmatrix}2x^2+16\\ 20\end{bmatrix}\quad\begin{bmatrix}48\\ 12x\end{bmatrix}$$

$3x+8=20$

$3x=12$

$x=12$

$x=4$

$2x+10x=48$

$12x=48$

$x=\frac{48}{12}$

$x=4$



Transpose of A Matrix

Matrix = A

Transpose of A = A^T

↳ Rows & coloumns interchange

$$A = \begin{bmatrix} 1 & 7 & -3 \\ 2 & 4 & 9 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \end{matrix}$$

$$A^T = \begin{bmatrix} 1 & 2 \\ 7 & 4 \\ -3 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 7 \\ 10 & 13 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 4 & 10 \\ 7 & 13 \end{bmatrix}$$



Symmetric & Skew-Symmetric Matrix

Symmetric

$A^T = A$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 5 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 0 & 5 \end{bmatrix}$$

$A = A^T$

Skew-Symmetric

$A^T = -A$

$A = -A^T$

$$A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$$

$$-A^T = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix} = A$$

$A = -A^T$

AI Board Booster: Student se Super Student

Ek aisa course jo AI ke use se aapki Board prep level-up kare

Outcomes- Aapko kya milega? (padhai mein badlao)

- **Faster revision:** Turn chapters into crisp **summaries, flashcards, and quizzes** in minutes.
- **Target weak areas:** AI **flags your common mistakes** and suggests exact topics to revisit.
- **Exam-style practice:** Convert **past papers into timed mocks** with stepwise solution.
- **Showcase work:** Build **8–10 mini projects** (notes generator, quiz bot, image-to-notes, etc.).
- **Repeatable study systems:** Unit tests & Boards ke liye **plug-and-play** checklists/templates.
- **Responsible AI use:** Kya **allowed** hai, kaise cite karna hai, and **over-reliance** avoid karna
- **Real-world edge: Prompting & automation** basics jo college/internships mein use karoge.
- **Recognized proof: PW Skills certificate** *(signed by Founder Alakh Pandey).*

Enroll Now →

For classes 9-12 (all boards)

2500 → 799 Parishram



Course at a glance

- **Live Online | Hinglish | Mentor:** Vishal Tiwari Sir
- **18 sessions × 3 hrs (Saturdays)**
- **Weekly Doubt Room:** Mon, 45–60 min
- **Start:** 13 Sept 2025
- **Fee:** ~~₹2500~~ → ₹799 for Uday & Parishram paid users *(coupon in tray)*

Limited seats - Offer ends soon

HW → (Community)

1) $A = \begin{bmatrix} 2 & 4 \\ 9 & 3 \\ 7 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 13 & 19 \\ 17 & 2 \\ 9 & -4 \end{bmatrix}$

(i) $A + B$

(ii) $A - B$

2) $C = \begin{bmatrix} 7 & -4 \\ 2 & 3 \end{bmatrix}$ $D = \begin{bmatrix} 19 & 23 \\ 14 & 19 \end{bmatrix}$

(i) C^T

(ii) $C + D$

(iii) $C - D$

3) $E = \begin{bmatrix} 7 & 3 & -4 \\ 2 & 9 & 4 \\ 10 & 11 & 2 \end{bmatrix}$ $F = \begin{bmatrix} 19 & 2 & 18 \\ 5 & -9 & 2 \\ 7 & 3 & 7 \end{bmatrix}$

(i) E^T

(ii) $E + F$

(iii) $E - F$

THANK YOU